Chapter 5 — Solids and Fluid Dynamics

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Types of solids and deformation

Q.1 Distinguish between crystalline, amorphous and polymeric solids.

Crystalline solids have a regular, repeating three-dimensional arrangement of atoms or molecules and a sharp melting point, for example salt, diamond and metals. Amorphous solids have no long-range order; their particles are arranged irregularly, and they soften over a range of temperature, for example glass and rubber. Polymeric solids consist of long chain molecules; they may be natural such as cellulose and rubber, or synthetic such as polyethylene and nylon.

Deformation in one dimension may be tensile, when the length increases under a pulling force, or compressive, when the length decreases under a pushing force. Within the elastic limit the deformation is reversible, while beyond it the deformation becomes permanent.

Type	Arrangement	Melting behaviour	Examples
Crystalline	Regular repeating lattice	Sharp melting point	Salt, diamond, metals
Amorphous	Irregular, no long-range order	Softens over a range	Glass, rubber
Polymeric	Long chain molecules	Varies with polymer	Cellulose, nylon, polyethylene

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Important Conceptual Short Questions — Topic 1

1. Why does a crystalline solid have a sharp melting point?

All particles are held in a regular lattice with nearly equal binding energies. At a definite temperature the whole lattice breaks down together, so melting occurs sharply.

2. How does an amorphous solid differ from a crystalline solid in melting?

An amorphous solid has irregular bonding, so different regions soften at different temperatures. It therefore softens over a range instead of melting at a sharp point.

3. Why are polymeric solids often light and flexible?

They are made of long chain molecules that can slide or uncoil under stress. The chains are not packed in a rigid lattice, so many polymers are light and flexible.

Topic 2 — Stress, strain and Young's modulus

Q.2 Define stress, strain and Young's modulus and describe Searle's apparatus for determining Young's modulus of a wire.

Stress is the restoring force per unit area, $\sigma = F/A$, measured in N m^{-2} or pascal. Strain is the fractional change in length, $\varepsilon = \Delta L/L_0$, and has no unit. Within the proportional limit, stress is proportional to strain, and the constant ratio is Young's modulus: $Y = \text{stress/strain} = (F/A)/(\Delta L/L_0) = FL_0/(A\Delta L)$.

In Searle's apparatus two identical wires are suspended from the same rigid support. One wire carries a fixed load and the other carries the variable load whose extension is measured. A spirit level and micrometer arrangement measures the small extension of the experimental wire relative to the reference wire, eliminating errors due to yielding of the support or temperature changes. A graph of load against extension gives a straight line, and Y is calculated from its slope.

Important Conceptual Short Questions — Topic 2

1. Why is strain a dimensionless quantity?

Strain is the ratio of extension to original length, $\Delta L/L_0$. Both are lengths, so the units cancel and strain has no dimensions.

2. How does Searle's apparatus remove errors due to temperature and support yielding?

Both wires are identical and suspended from the same support. Temperature changes or support movement affect both wires equally, so the relative extension measured is due only to the added load.

3. Why is Young's modulus a property of the material and not of the wire?

Young's modulus is the ratio of stress to strain, which removes the effects of the wire's length and cross-sectional area. It therefore depends only on the material.

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Topic 3 — Stress–strain curve and strain energy

Q.3 Explain the important points on a stress–strain curve for a ductile material and derive the elastic potential energy stored in a stretched wire.

The curve begins with a straight region where stress is proportional to strain; the end of this region is the proportional limit. Up to the elastic limit the material returns to its original length when the load is removed. Beyond the elastic limit, or yield point, plastic deformation begins. The stress then reaches a maximum called the ultimate tensile strength, after which the wire necks and finally breaks at the fracture point.

The work done in stretching a wire is stored as elastic potential energy. Since the force increases linearly from 0 to F, the average force is F/2, so the energy stored

is $E = \frac{1}{2}F\Delta L$. Using $F = k\Delta L$, this becomes $E = \frac{1}{2}k(\Delta L)^2$. It is equal to the area under the force–extension graph.

Important Conceptual Short Questions — Topic 3

1. Why is the area under a force–extension graph equal to the energy stored?

Each small strip $F\,\Delta x$ represents the work done in a small extension. Adding all strips gives the total work, which is stored as elastic potential energy, so the area equals the energy.

2. How does elastic deformation differ from plastic deformation?

In elastic deformation the material returns to its original shape when the load is removed. In plastic deformation the change is permanent because the material has been stressed beyond its elastic limit.

3. Why does a ductile material show a yield point?

Beyond the elastic limit the layers of atoms begin to slip permanently. A small increase in stress then produces a large strain, which appears as the yield point on the curve.

Topic 4 — Archimedes' principle and floatation

Q.4 State Archimedes' principle and explain the floatation of ships.

Archimedes' principle states that a body wholly or partly immersed in a fluid experiences an upthrust equal to the weight of the fluid displaced. The upthrust is $U = \rho g V$, where ρ is the fluid density and V is the displaced volume. A body floats when the upthrust equals its weight; it sinks when its weight is greater than the maximum upthrust.

A steel ship floats although steel is denser than water because its hollow shape displaces a large volume of water. The weight of this displaced water equals the weight of the ship, so the upthrust balances the weight. A solid block of steel displaces only its own small volume, so the upthrust is less than its weight and it sinks.

Important Conceptual Short Questions — Topic 4

1. Why does a steel ship float while a steel block sinks?

The ship's hollow shape displaces a large volume of water whose weight equals the ship's weight. The solid block displaces only a small volume, so the upthrust is less than its weight.

2. How does Archimedes' principle explain the feeling of lightness in water?

The water exerts an upward upthrust equal to the weight of water displaced by the body. This upthrust opposes the weight, so the apparent weight is reduced.

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3. Why does a floating body displace fluid equal to its own weight?

For equilibrium the upward upthrust must balance the downward weight. Since upthrust equals the weight of displaced fluid, the displaced fluid must have the same weight as the body.

Topic 5 — Equation of continuity and Bernoulli's equation

Q.5 Derive the equation of continuity and Bernoulli's equation for an ideal fluid.

For steady flow of an incompressible fluid, the mass entering a tube per second must equal the mass leaving it. If the cross-sectional areas are A_1 and A_2 and the speeds are v_1 and v_2, then $\rho A_1 v_1 = \rho A_2 v_2$, so $A_1 v_1 = A_2 v_2$. This is the equation of continuity and expresses conservation of mass.

Bernoulli's equation applies to steady, incompressible, non-viscous flow along a streamline. It states that $P + \frac{1}{2}\rho v^2 + \rho g h$ is constant. It is a statement of conservation of energy: pressure energy, kinetic energy and potential energy per unit volume can change into one another. For horizontal flow, $P + \frac{1}{2}\rho v^2 =$ constant, so where the speed is high the pressure is low.

Important Conceptual Short Questions — Topic 5

1. Why does squeezing the end of a rubber pipe increase the speed of water?

The equation of continuity requires Av to remain constant. When the area A is reduced, the speed v must increase so that the same volume flows per second.

2. Why is the pressure lower where a fluid moves faster?

In horizontal flow, Bernoulli's equation requires $P + \frac{1}{2}\rho v^2$ to be constant. An increase in kinetic energy per unit volume must be balanced by a decrease in pressure.

3. How does the equation of continuity express conservation of mass?

For an incompressible fluid, the volume entering a pipe per second must equal the volume leaving it. Since density is constant, the mass flow rate is also constant.

Topic 6 — Applications of Bernoulli's equation

Q.6 Explain any four applications of Bernoulli's equation.

An aeroplane wing is shaped so that air moves faster over the upper surface than the lower surface. The pressure above is lower, producing lift. A swinging ball curves because the spin makes the air speed greater on one side, so the pressure difference pushes the ball sideways. A filter pump and a carburettor use a fast stream of fluid to produce low pressure that draws in another fluid. A paint sprayer uses fast air to lower pressure and draw paint up the tube.

The Venturi relation shows that a constriction in a horizontal pipe increases speed and lowers pressure, and is used to measure flow speed. Torricelli's theorem gives

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the speed of efflux from a hole at depth h as $v = \sqrt{(2gh)}$, which is the speed a body would acquire falling freely through height h.

Important Conceptual Short Questions — Topic 6

1. How does the shape of an aeroplane wing produce lift?

Air travels faster over the curved upper surface, so the pressure there is lower than below the wing. The pressure difference produces an upward lift force.

2. Why does a spinning ball swing in flight?

The spin makes the air speed greater on one side of the ball and smaller on the other. By Bernoulli's principle the pressure is lower on the faster side, so the ball is pushed sideways.

3. How does Torricelli's theorem relate efflux speed to depth?

Applying Bernoulli's equation between the surface and the hole gives $v = \sqrt{(2gh)}$. The speed of the emerging liquid is the same as that of a body falling freely through the depth h.

Topic 7 — Viscosity, terminal velocity and superfluids

Q.7 Explain viscous drag, Stokes' law and terminal velocity, and describe superfluids.

Viscosity is the internal friction of a fluid that opposes relative motion between its layers. For a small sphere of radius r moving with speed v through a fluid of viscosity η, Stokes' law gives the drag force $F = 6\pi\eta rv$. A falling droplet accelerates until the drag plus upthrust equals its weight; then the net force is zero and it falls with constant terminal velocity. For a sphere of density ρ falling through a fluid of negligible density, $v_t = 2gr^2\rho/9\eta$.

Superfluids are liquids that flow with zero viscosity below a critical temperature, such as liquid helium below about 2 K. They can creep over the walls of containers and, once stirred, their vortices keep spinning indefinitely because there is no viscous dissipation.

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Important Conceptual Short Questions — Topic 7

1. Why does a falling droplet reach a terminal velocity?

As speed increases, the viscous drag increases. When drag plus upthrust equals the weight, the net force becomes zero, so acceleration stops and the droplet falls at constant speed.

2. Why is the acceleration zero at terminal velocity?

At terminal velocity the viscous drag plus upthrust exactly balance the weight, so the net force is zero. By Newton's second law the acceleration is then zero and the velocity remains constant.

3. Why can a superfluid creep over the walls of a container?

A superfluid has zero viscosity, so there is no internal friction to oppose flow. It can therefore move as a thin film over surfaces and even climb the walls of a container.

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